**LUMEXA**
Space-themed coding education for K–12

AGES 12–18 | BEGINNER FRIENDLY | NO PRIOR CODING EXPERIENCE NEEDED

CURRICULUM OVERVIEW

# Python Foundations

## Build 9 Real Python Projects

A complete, hands-on journey into text-based Python. Students start from the very first line of code, then learn to solve problems, structure and debug programs, work with files and external libraries, and explore the tools used in modern Python development.




**9** REAL PYTHON PROJECTS
From a calculator to a data capstone

**4** TESTS ALONG THE WAY
3 course tests + 1 final test

| 3 | 24 | 28 | 12–18 |
|---|---|---|---|
| Courses | Lessons | Sessions in total | Age range |

**WHO IT IS FOR**
Beginner friendly — no prior coding experience needed

**BEFORE YOU START**
No coding experience needed

## LEARNING PROGRESSION

**01 Python Programming Essentials**
Write independent Python programs from the very first line.

**02 Problem Solving & Program Design**
Plan, debug, handle errors and build with files and classes.

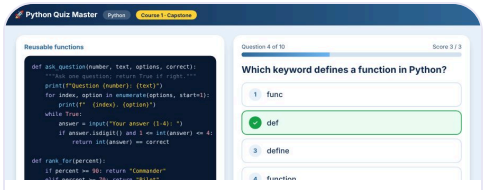
**03 Python Tools & Data Foundations**
Use real tools and data: notebooks, NumPy, Pandas and APIs.

### Learn to think like a programmer
Students break down problems, design solutions and approach unfamiliar challenges independently, not just copy code.

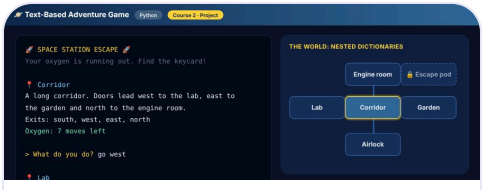
### A portfolio, not just a certificate
Students finish with nine real projects they can build, demonstrate and add to their portfolios.

### Tested, so progress is real
Every course ends with a test, and a final test closes the program, so progress is proven, not assumed.

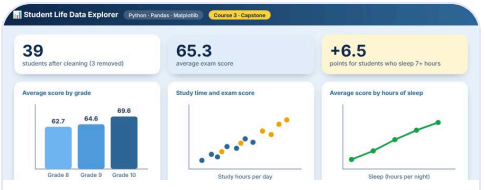
## SAMPLE HANDS-ON PROJECTS STUDENTS BUILD




**01 Python Quiz Master**
Python · Functions




**02 Text-Based Adventure Game**
Python · Dictionaries




**03 Student Life Data Explorer**
Python · Pandas · Matplotlib

COURSE 01 · LESSONS 1–8

# 01 Python Programming Essentials

Write your first real Python programs, from scratch

Python 3 | Python editor | Control flow | Collections | Functions

3 REAL PROJECTS

1 COURSE TEST

## OVERVIEW

No coding experience needed. Students learn how to write, understand and run Python programs, and gradually combine variables, decisions, loops, strings, collections and functions into interactive applications.

## PORTFOLIO PROJECTS

- Smart Calculator
- Number Guessing Game
- Python Quiz Master

## WHY IT MATTERS

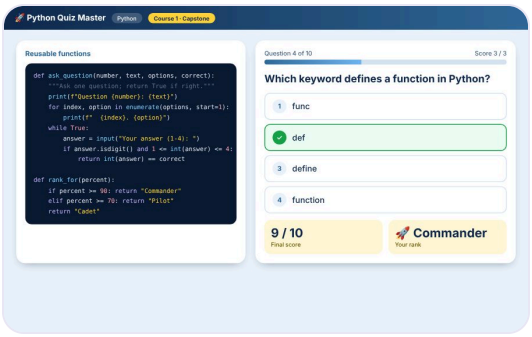
Every programmer starts here. Fluency with syntax, data types, conditions, loops and functions is the foundation for everything that follows, from games and websites to data science and AI.

**8** LESSONS | **3** PROJECTS | **1** COURSE TEST

## LESSON BREAKDOWN

1. Python Setup and Your First Program
2. Variables, Data Types, and Values
3. User Input, Output, and Operators
4. Making Decisions with Conditional Logic
5. Repetition with Python Loops
6. Working with Strings
7. Working with Python Collections
8. Creating Reusable Code with Functions

8 LESSONS → TEST UNLOCKS




SAMPLE HANDS-ON PROJECT

### Python Quiz Master

Python · Functions

A quiz application built from reusable functions: it asks questions, checks the answers, keeps score and awards a rank.

COURSE TEST

### Python Programming Essentials — Course Test

Checks what you built and learned in this course. Taken after lesson 8, before the next course begins.

30 multiple-choice | 10 true / false | 40 marks

COURSE 02 · LESSONS 9–16

# 02 Problem Solving & Program Design

Think, solve, structure and build like a programmer

Python 3 | Algorithms | Debugging | Files · JSON · CSV | OOP

3 REAL PROJECTS

1 COURSE TEST

## OVERVIEW

Students move beyond single statements and learn to approach problems systematically: designing solutions, organising larger programs, finding and fixing errors, handling the unexpected and working with persistent data.

## PORTFOLIO PROJECTS

- Text-Based Adventure Game
- Personal Expense Tracker
- Library Management System

## WHY IT MATTERS

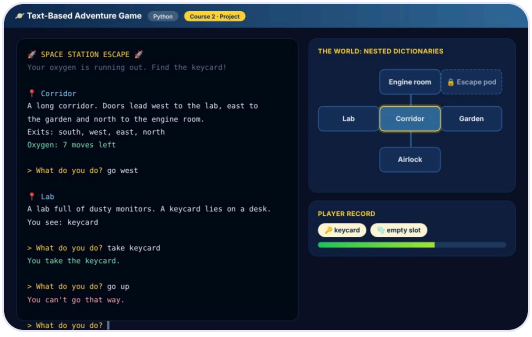
Writing code is only half of programming. Planning, testing, debugging and organising code into functions and classes is what turns a script into software that other people can trust.

| 8 | 3 | 1 |
|---|---|---|
| LESSONS | PROJECTS | COURSE TEST |

## LESSON BREAKDOWN

1. Computational Thinking & Problem Decomposition
2. Functions in Depth & Modular Programming
3. Advanced Collections & Nested Data Structures
4. Algorithms & Problem-Solving Patterns
5. Debugging, Testing & Fixing Problems
6. Error Handling & Input Validation
7. Working with Files & Structured Data
8. Introduction to Object-Oriented Programming

8 LESSONS — TEST UNLOCKS




SAMPLE HANDS-ON PROJECT

### Text-Based Adventure Game

Python · Dictionaries

A text adventure: the world is a nested dictionary, the player is a record, and small functions run the game.

COURSE TEST

### Problem Solving & Program Design — Course Test

Checks what you built and learned in this course. Taken after lesson 8, before the next course begins.

30 multiple-choice | 10 true / false | 40 marks

COURSE 03 · LESSONS 17–24

# 03 Python Tools & Data Foundations

Prepare for real-world Python development

VS Code · Git · Jupyter · NumPy · Pandas · Matplotlib

3 REAL PROJECTS

1 COURSE TEST

## OVERVIEW

Students explore the tools professionals use: development environments, libraries and packages, Jupyter Notebooks, NumPy, Pandas and APIs. They take their first steps with real data and learn to document and share a finished project.

## PORTFOLIO PROJECTS

- Dataset Explorer
- Live Data Explorer
- Student Life Data Explorer

## WHY IT MATTERS

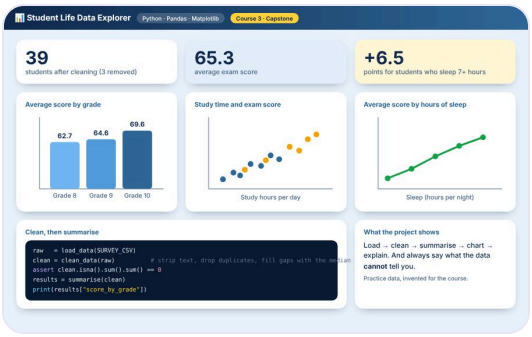
This course is the bridge to specialisation. It introduces the workflows and libraries behind data science and AI, so students arrive at those programs with confidence in their tools.

**8** LESSONS | **3** PROJECTS | **1** COURSE TEST

## LESSON BREAKDOWN

1. Python Development Environment & Workflow
2. Python Modules, Libraries & Packages
3. Introduction to Jupyter Notebooks
4. Introduction to NumPy
5. Introduction to Pandas & DataFrames
6. Data Cleaning, Summaries & Basic Visualization
7. Working with APIs & JSON Data
8. Python Capstone: Build, Document & Share

8 LESSONS → TEST UNLOCKS




SAMPLE HANDS-ON PROJECT

### Student Life Data Explorer

Python · Pandas · Matplotlib

A capstone data project: load a survey, clean it, summarise it with Pandas and tell the story in three charts.

COURSE TEST

### Python Tools & Data Foundations — Course Test

Checks what you built and learned in this course. Taken after lesson 8, before the next course begins.

30 multiple-choice · 10 true / false · 40 marks

HOW STUDENTS ARE TESTED · 4 TESTS

# Level-Up Checkpoints

Every course ends with a test of both ideas and practical skills, and a final test closes the program with questions and an independent Python project.



THE CHECKPOINT PATH — EVERY COURSE ENDS WITH A TEST, THEN ONE FINAL




COURSE TEST · every 8 lessons

## Show what you built and learned

After every 8 lessons a course test checks the ideas and skills of that course, so each next course starts on solid ground.

| | |
|---|---|
| 30 | multiple-choice questions — choose one of four answers |
| 10 | true / false questions |
| 40 | marks in total — 1 mark for every question |

FINAL TEST · the finish line

## One final test on the whole curriculum

One final test across the whole program: questions on Python, problem solving and tools, plus a practical programming task.

| | |
|---|---|
| 25 | multiple-choice questions |
| 5 | true / false questions |
| 20 | marks for one small practical project (about 15 minutes) that your teacher marks |

## A TASTE OF A TEST

SAMPLE · MULTIPLE-CHOICE

**Which Python function shows text on the screen?**

- A input()
- B print() ✓ CORRECT
- C range()
- D len()

SAMPLE · TRUE / FALSE

**A variable created inside a function can be used anywhere in the program.**

True | False ✓

Examples in the style of a real test — written for this document, not taken from one.

## AFTER EVERY TEST

**Saved as you go**
Answers save automatically while a student works, so nothing is lost if the connection drops.

**Checked by the teacher**
The teacher reviews the report and shares the result with the student.

**Fair for everyone**
Questions and answer options appear in a different order for every student.

# The Complete Journey

Across 3 courses, 24 lessons and 4 tests, students progress from writing their first Python statement to building structured applications and exploring real-world data, finishing with nine projects for their portfolio.

## 3 COURSES · 24 LESSONS

**01 Python Programming Essentials**
Smart Calculator · Number Guessing Game · Python Quiz Master
8 lessons · + course test

**02 Problem Solving & Program Design**
Text-Based Adventure Game · Personal Expense Tracker · Library Management System
8 lessons · + course test

**03 Python Tools & Data Foundations**
Dataset Explorer · Live Data Explorer · Student Life Data Explorer
8 lessons · + course test

## HOW STUDENTS ARE TESTED · 4 TESTS

COURSE TEST 1
Python Programming Essentials

COURSE TEST 2
Problem Solving & Program Design

COURSE TEST 3
Python Tools & Data Foundations

FINAL TEST
**One final test on the whole curriculum**
One final test across the whole program: questions on Python, problem solving and tools, plus a practical programming task.

## BY THE END, STUDENTS CAN

- Write independent Python programs using variables, data structures, functions, conditions, loops and objects.
- Break down problems, design algorithms, debug, handle errors and work with files to build complete applications.
- Work with real Python tools and data: libraries, Jupyter Notebooks, NumPy, Pandas, APIs and basic charts.

**Ready to start building?**
All lessons and projects are taught live by an expert Lumexa instructor.

Lumexa Python Foundations · Ages 12–18